Binary Outcomes: LPM, Logit, and Probit

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ECO 351 – Introduction to Econometrics

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Recall: Reporting outcome units

From Lecture 23

A ten-cent tax increase was translated into a change in traffic deaths per 10,000 residents.

$$\Delta \widehat{Y} = (-0.640)(0.10) = -0.064$$

Does this mean 0.064 fewer total deaths in the state? What additional information would be needed to convert the rate change to a count?

Why outcome units matter before changing models

Answer

The estimate is 0.064 fewer deaths per 10,000 residents, not 0.064 fewer total deaths. Converting the rate to an expected count also requires the state population.

What does a regression coefficient mean when the outcome takes only the values zero and one?

Mortgage denial as a zero-one outcome

The Boston HMDA dataset contains 2,380 mortgage applications. Let $Y = 1$ indicate denial and $Y = 0$ approval.

```
. tabulate deny
       deny |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 |      2,095       88.03       88.03
          1 |        285       11.97      100.00
------------+-----------------------------------
      Total |      2,380      100.00
```

The average of a zero-one outcome is its sample proportion: here $285/2380 \approx 0.120$.

PIR is the proposed payment-to-income ratio. PIR 0.30 means payments equal 30% of income.

The conditional mean is a probability

$$E(Y \mid X) = 1 \cdot P(Y = 1 \mid X) + 0 \cdot P(Y = 0 \mid X)$$

$$E(Y \mid X) = P(Y = 1 \mid X)$$

For a binary outcome, modeling the conditional mean is therefore modeling the probability of the event.

We write this probability as $p(X)$. In a linear probability model,

$$p(X) = \beta_0 + \beta_1 X.$$

The model approximates how denial probability varies with PIR.

Stata: the linear probability model

```
. regress deny pir, vce(robust)
Linear regression                               Number of obs     =      2,380
------------------------------------------------------------------------------
             |               Robust
        deny | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         pir |   .6035349   .0984826     6.13   0.000     .4104144    .7966555
       _cons |  -.0799096   .0319666    -2.50   0.012    -.1425949   -.0172243
------------------------------------------------------------------------------
```

The fitted slope is about 0.604 probability units per one-unit PIR increase. A 0.10 increase corresponds to about 6.04 percentage points.

Probabilities and percentage points

Using the rounded fitted line,

$$\widehat{p}(x) = -0.080 + 0.604x,$$

$$\widehat{p}(0.30) = 0.1012, \qquad \widehat{p}(0.40) = 0.1616.$$

The difference is 0.0604, or **6.04 percentage points**.

Relative to the starting probability, the increase is $0.0604/0.1012 \approx 59.7\%$. This is a different comparison.

Exercise: a smaller payment increase

$$\widehat{p} = -.080 + .604PIR$$

Your task

Using the rounded equation, compare PIR 0.25 with 0.30. Calculate both probabilities and their difference. Explain the units.

Debrief: the change is in probability units

$$\widehat{p}(0.25) = 0.0710, \quad \widehat{p}(0.30) = 0.1012$$

$$\Delta\widehat{p} = 0.0302 = 3.02 \text{ percentage points}$$

The PIR increases by 0.05, not by five whole units.

The straight-line restriction

The linear probability model gives the same slope at every PIR. It can also produce predictions outside $[0, 1]$.

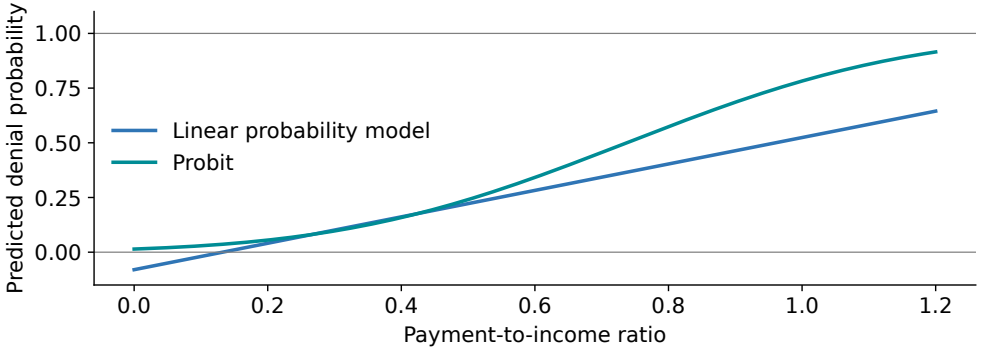
$$\widehat{p}(0) = -0.080.$$

For a binary outcome with conditional probability $p(X)$,

$$\text{Var}(Y \mid X) = p(X)[1 - p(X)].$$

Robust standard errors address the variance issue. They leave the fitted line and its probability-range problem unchanged.

A bounded probability curve



A probability curve flattens near zero and one. Its slope need not be constant across starting PIR values.

The probit model

Let Φ be the standard normal cumulative distribution function. Probit specifies

$$p(X) = \Phi(\eta), \qquad \eta = \beta_0 + \beta_1 X.$$

The **index** η can take any real value. Applying Φ gives a probability between zero and one.

To compare two PIR values, calculate their two indices, convert each to a probability, then subtract the probabilities.

Stata: probit coefficient output

```
. probit deny pir, vce(robust)
Probit regression                                       Number of obs =  2,380
------------------------------------------------------------------------------
             |               Robust
        deny | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         pir |   2.967907   .4653217     6.38   0.000     2.055893    3.879921
       _cons |  -2.194159   .1649761   -13.30   0.000    -2.517506   -1.870812
------------------------------------------------------------------------------
Note: 0 failures and 1 success completely determined.
```

The fitted index is $-2.194159 + 2.967907\,PIR$.

The PIR coefficient changes the index. A probability comparison still requires the CDF.

Stata: probabilities at three PIR values

```
. margins, at(pir=(0.30 0.40 0.50))
Adjusted predictions                                      Number of obs = 2,380
------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         _at |
          1  |   .0961532   .0069017    13.93   0.000      .082626    .1096803
          2  |   .1569683   .0101166    15.52   0.000     .1371402    .1767965
          3  |   .2387884    .024485     9.75   0.000     .1907987    .2867781
------------------------------------------------------------------------------
```

Rows 1, 2, and 3 correspond to PIR 0.30, 0.40, and 0.50.

Exercise: equal PIR increases

Using the preceding probit output:

1. Calculate the probability change from PIR 0.30 to 0.40.
2. Repeat for 0.40 to 0.50.
3. Express both in percentage points and explain why they differ.

A colleague instead multiplies 2.967907 by 0.10. What does that calculation change?

Debrief: two finite probability changes

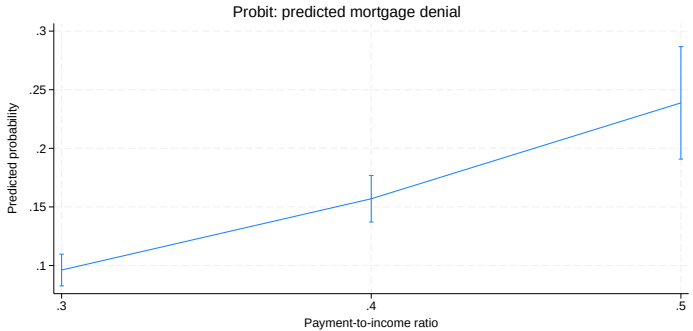
PIR change	Probability difference	Percentage points
0.30 to 0.40	$0.1569683 - 0.0961532$	6.08
0.40 to 0.50	$0.2387884 - 0.1569683$	8.18

The second comparison crosses a steeper part of the fitted curve.

The change in the index is the same for each increment. The corresponding probability changes differ.

The predicted probabilities and their intervals

`marginsplot` displays the preceding predictions.



Bars show 95% intervals for each fitted probability. A probability-difference interval requires the uncertainty of the difference.

The logit model

Logit uses the logistic CDF, written Λ:

$$p(X) = \Lambda(\eta) = \frac{1}{1 + \exp(-\eta)}, \qquad \eta = \beta_0 + \beta_1 X.$$

Like probit, it gives bounded probabilities with a slope that can vary across starting values.

The two models put their indices on different scales. Compare their predicted probabilities and probability changes.

Stata: logit coefficient output

```
. logit deny pir, vce(robust)
Logistic regression                                     Number of obs =  2,380
------------------------------------------------------------------------------
             |               Robust
        deny | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         pir |   5.884498   .9999333     5.88   0.000     3.924664    7.844331
       _cons |  -4.028432   .3589036   -11.22   0.000     -4.73187   -3.324994
------------------------------------------------------------------------------
```

The PIR coefficient is about 5.884, versus 2.968 in probit. This does not imply a probability effect almost twice as large.

A complete binary-outcome comparison

For a probability comparison, specify the outcome coding, starting and ending regressor values, and any controls held fixed. Calculate the two probabilities before reporting their difference.

Why is “the probit coefficient is 2.97, so denial probability rises by 2.97” incorrect?

Next class: how the model estimates its coefficients, how to obtain uncertainty for a probability difference, and a cumulative application review.